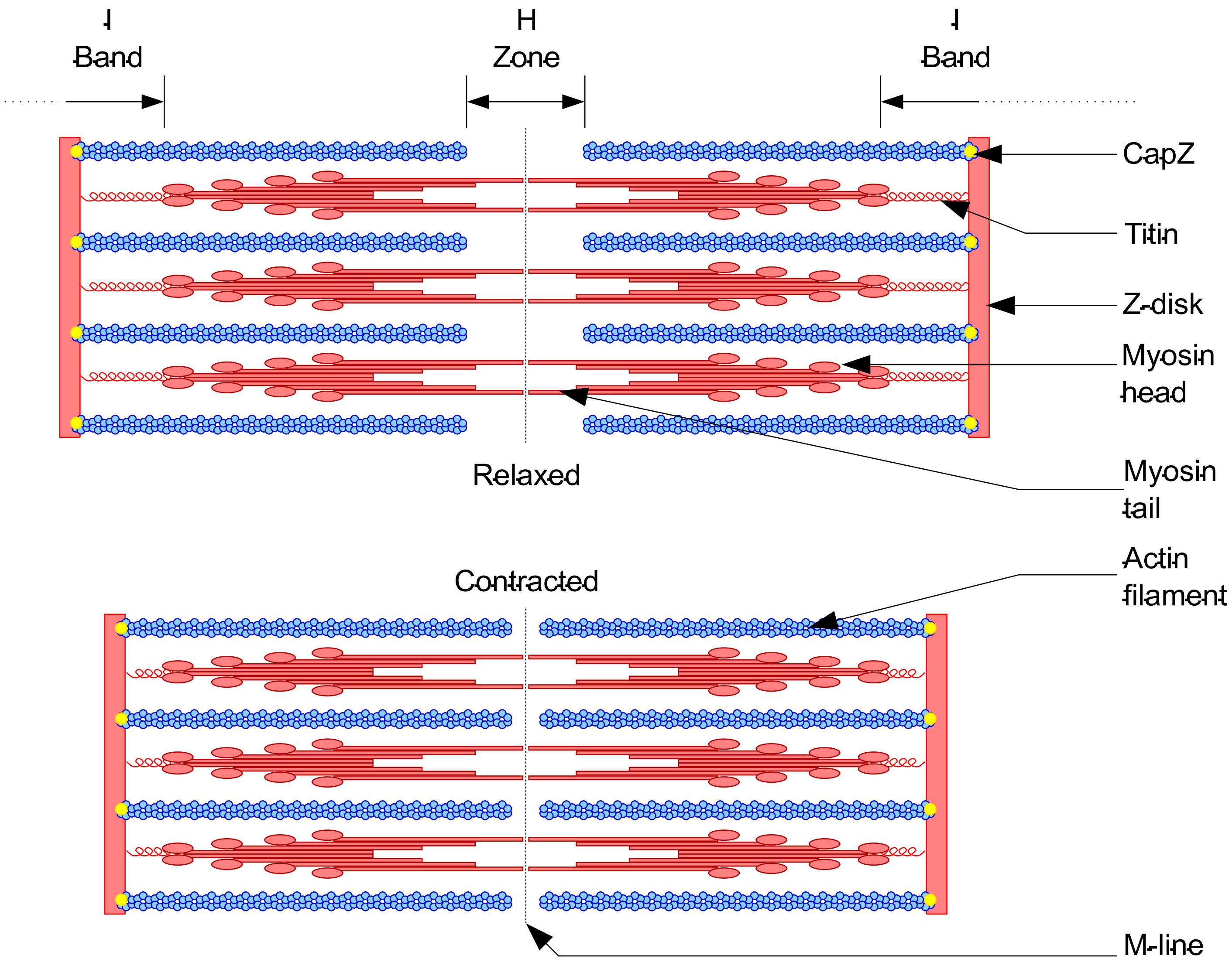
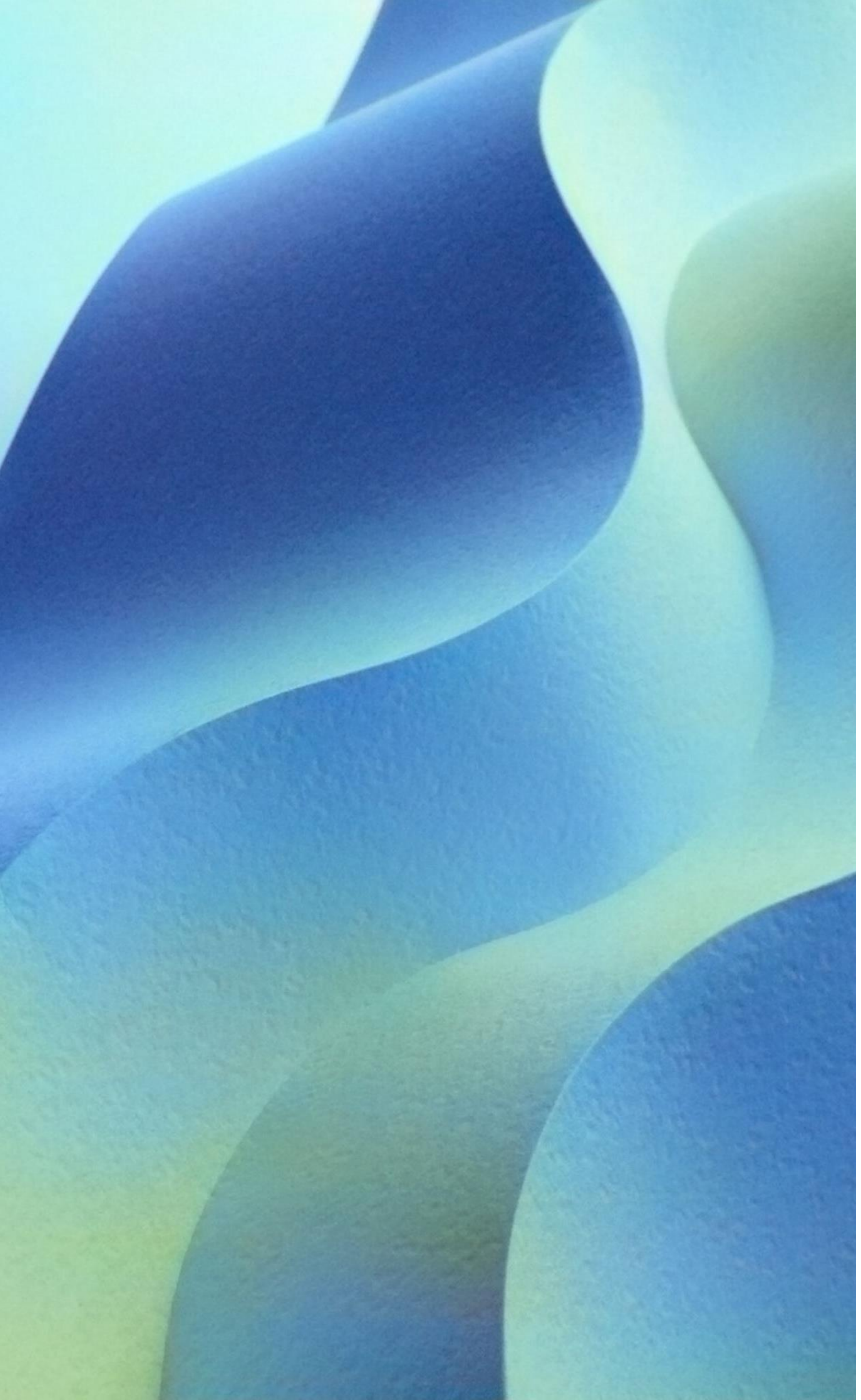
Muscle Tissue II

BIOL2251

Chapter 10

Sections 10.4 – 10.5





Pre-Lecture Quiz

Test what you *know*!

10 minutes for first attempt

Password = **titin**

No notes – just what is stored in your brain

Resting Membrane Potential

Resting Membrane Potential

- **Instructions**
  - 5 minutes
  - Individual work
  - On paper, write:
    - **(1) What the resting membrane potential value is**
    - **(2) Which ions are most important and where they are concentrated**
    - **(3) What it means for the inside of the cell to be 'negative' at rest**
  - Review as a class
- **Bloom’s taxonomy = Remember, Understand**

Resting Membrane Potential

- RMP of a muscle fiber = approximately -85 mV

Ion	Higher Concentration	Reason
Na+	Outside the cell	Na-K pump moves Na^+ out; some Na^+ leaks out through leak channels; membrane relatively impermeable to Na^+ at rest
K+	Inside the cell	Na-K pump moves K^+ in; K^+ leaks out through leak channels
Proteins (Large anions)	Inside the cell	Too large to cross the membrane; contribute to negative interior charge

- Resting because the cell is not generating a signal and the cell is polarized there is difference in charged across the membrane
  - Inside is negative because of net movement of K+ ions out of the cell and the retention of negatively-charged proteins inside the cell

Action Potential

Step Sequencing

Action Potential – Step Sequencing

- **Instructions**
  - 8 minutes
  - Work in pairs
  - **Arrange the scrambled steps of an action potential in the correct order**
  - Debrief as a class
- **Bloom’s taxonomy = Understand, Apply**

Action Potential – Step Sequencing

A. Membrane repolarizes as K^+ channels open and K^+ flows out of the cell.

B. Voltage-gated Na^+ channels open; Na^+ rushes into the cell down its electrochemical gradient.

C. Membrane briefly hyperpolarizes before returning to resting potential.

D. A stimulus depolarizes the membrane to threshold.

E. Voltage-gated Na^+ channels inactivate; Na^+ entry stops.

F. The membrane potential rapidly rises toward +30 mV.

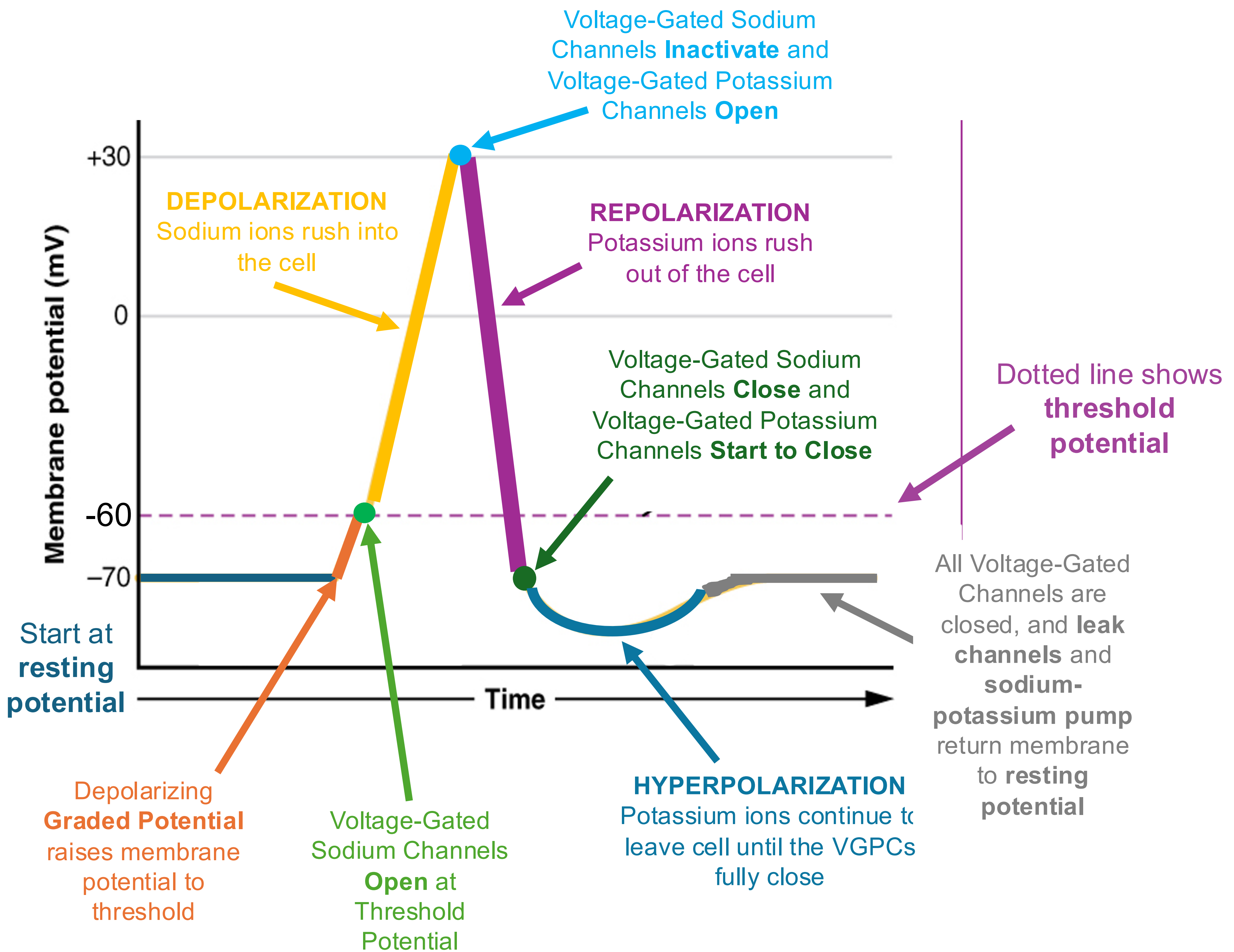
G. The Na^+/K^+ pump restores original ion concentrations over time.

H. K^+ channels close slowly; some K^+ has moved out beyond what is needed for repolarization.

Action Potential – Step Sequencing

Step	Letter	Event
1	D	A stimulus depolarizes the membrane to threshold
2	B	Voltage-gated Na^+ channels open; Na^+ rushes into the cell down its electrochemical gradient
3	F	The membrane potential rapidly rises toward +30 mV
4	E	Voltage-gated Na^+ channels inactivate; Na^+ entry stops
5	A	Membrane repolarizes as K^+ channels open and K^+ flows out of the cell
6	C	Membrane briefly hyperpolarizes before returning to resting potential
7	H	K^+ channels close slowly; some K^+ has moved out beyond what is needed for repolarization
8	G	The Na^+/K^+ pump restores original ion concentrations over time.

Action Potential



Action Potential – Step Sequencing

- A drug blocks all voltage-gated Na^+ channels in a neuron. The resting membrane potential is normal. What happens when the neuron receives a stimulus, and why?
- No action potential would be generated
  - With voltage-gated Na^+ channels blocked, the membrane can be depolarized to threshold but cannot produce the rapid Na^+ influx that drives the depolarization phase of the action potential
  - The stimulus would produce only a local, graded depolarization that fades — not a full action potential
  - This is the mechanism of local anesthetics (e.g., lidocaine)

Muscle Contraction Process

Categorize and Sequence

Muscle Contraction Process

- **Instructions**
  - 20 minutes
  - Work in small groups (4-6 people)
  - **Step 1: Sort the 16 events into the correct phase — Events at the NMJ, Excitation-Contraction Coupling, or Contraction Cycle**
  - **Step 2: Put the events in the correct order within each phase**
  - **Step 3: Next to each event, write the key structure, protein, or ion responsible**
  - We will break down each phase at the end
- **Bloom's taxonomy = Apply, Analyze**

Events at the NMJ

Order	Event	Key Structure/Protein/Ion
1	B — Action potential arrives at axon terminal	Voltage-gated channels (neuron)
2	G — Ca^{2+} enters axon terminal	Ca^{2+}; voltage-gated Ca^{2+} channels
3	D — ACh released into synaptic cleft by exocytosis	Acetylcholine (ACh)
4	J — ACh binds motor end plate receptors; membrane depolarizes	ACh; ACh receptors; Na^+
5	F — Action potential generated and travels along sarcolemma	Na^+; voltage-gated Na^+ channels

Excitation-Contraction Coupling

Order	Event	Key Structure/Protein/Ion
1	I — Action potential travels deep into muscle fiber	T-tubules; sarcolemma
2	E — Ca^{2+} floods the sarcoplasm from its storage site	Ca^{2+}; sarcoplasmic reticulum
3	L — Ca^{2+} binds regulatory protein on thin filament	Ca^{2+}; troponin
4	H — Tropomyosin shifts, unblocking myosin binding sites	Tropomyosin; troponin
5	A — Myosin binding sites on actin become exposed	Actin

Contraction Cycle

Order	Event	Key Structure/Protein/Ion
1	C — Myosin heads bind actin; cross-bridges form	Myosin; actin
2	K — Power stroke; thin filament pulled toward M line; ADP + Pi released	Myosin; ADP; Pi
3	M — ATP binds to myosin head; myosin detaches from actin	ATP; myosin
4	N — ATP hydrolyzed; myosin head returns to cocked, high-energy position	ATP → ADP + Pi; myosin
5	O — Cocked myosin head re-binds actin at a new position	Myosin; actin
6	P — Cycle repeats as long as Ca^{2+} remains elevated and ATP is available	Ca^{2+}; ATP

Muscle Relaxation

Mirror and Contrast

Muscle Relaxation – Mirror and Contrast

- **Instructions**
  - 8 minutes
  - Think on your own for 2 minutes, then work with a partner
  - **For each contraction event listed, write the corresponding relaxation event**
  - Pairs will volunteer to help us review
- **Bloom's taxonomy = Understand, Apply**

Muscle Relaxation – Mirror and Contrast

- ACh is released into the synaptic cleft
- ACh binds to receptors; motor end plate depolarizes
- Action potential travels along sarcolemma and down T-tubules
- Ca^{2+} is released from the sarcoplasmic reticulum
- Ca^{2+} binds troponin; tropomyosin shifts; binding sites exposed
- Myosin heads bind actin; cross-bridges form and cycle

Muscle Relaxation – Mirror and Contrast

Contraction Event	Corresponding Relaxation Event
ACh released into synaptic cleft	Acetylcholinesterase breaks down ACh in the synaptic cleft; no further motor end plate stimulation
ACh binds receptors; depolarization	No new action potentials are generated at the motor end plate
Action potential travels along sarcolemma and T-tubules	No further action potentials propagate; SR Ca^{2+} release channels close
Ca^{2+} released from sarcoplasmic reticulum	Ca^{2+} pumps actively transport Ca^{2+} back into the SR against its concentration gradient
Ca^{2+} binds troponin; binding sites exposed	Ca^{2+} detaches from troponin; tropomyosin returns to its blocking position over myosin binding sites
Cross-bridges form and cycle	Myosin heads detach from actin; cross-bridge cycling stops; muscle passively returns to resting length

Length-Tension Relationship

Predict and Justify

Length-Tension Relationship

- **Instructions**
  - 8 minutes
  - **For each sarcomere length scenario, predict the relative tension produced (low, moderate, or optimal)**
    - **Justify your answer using what you know about filament overlap and cross-bridge formation**
  - Work individually for 4 minutes, then compare with your group
  - Groups will be selected at random to share their work
- **Bloom’s taxonomy = Analyze, Evaluate**

Length-Tension Relationship

- **Scenario 1:** A sarcomere is at its resting length. Thick and thin filaments overlap substantially, but thin filaments from opposite ends of the sarcomere do not yet reach the center or interfere with each other.
- **Scenario 2:** A sarcomere has been passively stretched to a very long length. There is minimal overlap between thick and thin filaments — the thin filaments barely reach the edges of the thick filament region.

Length-Tension Relationship

- **Scenario 3:** A sarcomere is compressed to a very short length. Thin filaments from both sides have slid so far inward that they now overlap each other in the center of the sarcomere, and thick filaments are beginning to butt up against the Z lines.
- **Scenario 4:** A sarcomere is at a slightly shortened length — there is good thick-thin filament overlap, but thin filaments are beginning to encroach slightly into the H zone from both sides.

Length-Tension Relationship

Scenario	Tension Level	Justification
1	Optimal	Maximum filament overlap → maximum number of cross-bridges can form simultaneously → greatest tension production
2	Low	Minimal filament overlap → very few myosin heads can reach actin binding sites → very few cross-bridges form → minimal tension
3	Low	Thin filaments from opposite sides interfere with each other and thick filaments compress against Z lines → cross-bridge geometry is disrupted → reduced and disorganized force production
4	Moderate	Good overlap but slight interference beginning → slightly fewer functional cross-bridges than optimal → tension is good but submaximal

Twitch Phases and Summation

Classification Challenge

Twitch Phases and Summation: Part 1

- **Instructions**
  - 5 minutes
  - **On your own, match descriptions to the correct twitch phase**
    - **Latent period, contraction phase, relaxation phase**
  - Compare your work with a partner
    - Debrief as a class
- **Bloom's taxonomy = Remember, Understand**

Twitch Phases and Summation: Part 1

1. Cross-bridges are actively cycling; tension in the muscle is rising
2. The action potential has arrived, Ca^{2+} is being released, and the regulatory proteins are shifting — but tension has not yet begun to rise
3. Ca^{2+} is being pumped back into the sarcoplasmic reticulum; tropomyosin returns to its blocking position; tension falls
4. This phase includes the time for action potential propagation down T-tubules and Ca^{2+} release from the SR
5. This is the longest phase of a single twitch under normal conditions

Twitch Phases and Summation: Part 1

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- Contraction Phase
- Latent Period
- Relaxation Phase
- Latent Period
- Relaxation Phase

Twitch Phases and Summation: Part 2

- **Instructions**
  - 5 minutes
  - **Classify each scenario (treppe, wave summation, incomplete tetanus, complete tetanus) and then rank all four from lowest tension (1) to highest tension (4)**
  - Compare your work with a partner
    - Debrief as a class
- **Bloom's taxonomy = Analyze**

Twitch Phases and Summation: Part 2

- **Scenario A:** A muscle receives a rapid series of stimuli with no rest between them. The muscle never has time to relax at all, producing a smooth, sustained plateau of maximum tension.
- **Scenario B:** A second stimulus arrives just before the muscle has fully relaxed from the first. The residual Ca^{2+} from the first twitch is still partially elevated when the second stimulus releases more Ca^{2+}, producing a slightly higher peak than a single twitch.
- **Scenario C:** A series of stimuli arrive at a moderate frequency — fast enough that the muscle only partially relaxes between each stimulus, producing a sustained but "wavy" tension that is higher than a single twitch but not smooth.
- **Scenario D:** The first few twitches in a series are delivered with full relaxation between them, but each successive twitch is slightly stronger than the last, producing a staircase increase in tension over the first several contractions.

Twitch Phases and Summation: Part 2

Scenario	Classification	Rank (tension)
A	Complete tetanus	4 — highest; no relaxation; continuous Ca^{2+} elevation; maximum sustained tension
B	Wave summation	2 — higher than single twitch; second stimulus on top of partial relaxation
C	Incomplete tetanus	3 — high and sustained but with oscillations; partial relaxation between stimuli
D	Treppe	1 — lowest of the four; full relaxation between twitches; modest staircase increase due to residual Ca^{2+} and muscle warming

"What Went Wrong?"

Diagnostic Case Studies

Diagnostic Case Studies

- **Instructions**
  - 12 minutes
  - Work in small groups (4-6 people)
  - For each case, students must identify:
    - **(1) where in the contraction/relaxation sequence the disruption occurs**
    - **(2) which specific protein or ion is affected**
      (3) whether the consequence is failure to contract, failure to relax, or altered tension production
  - Groups will be selected at random to share
- **Bloom’s taxonomy = Apply, Evaluate**

Diagnostic Case Studies

Case 1: A patient is treated with a drug that permanently blocks ACh receptors on the motor end plate. The motor neuron fires normally and ACh is released normally into the synaptic cleft. The patient develops progressive muscle paralysis — no voluntary movement is possible. Where does the sequence fail, and why?

- Failure point: **Neuromuscular junction** — specifically the motor end plate
- Disrupted structure: **ACh receptors** (nicotinic receptors on sarcolemma)
- Consequence: ACh is present in the cleft but cannot bind and activate ion channels → action potential not relayed to the motor end plate → no action potential generated on the sarcolemma → excitation-contraction coupling never initiated → no contraction despite normal neural activity

Diagnostic Case Studies

Case 2: A researcher exposes a muscle fiber to a toxin that permanently locks Ca^{2+} pumps in the SR membrane in an inactive state. The muscle contracts normally when stimulated. What happens when stimulation stops, and what is the consequence for the muscle?

- Failure point: **Relaxation phase** — Ca^{2+} removal from sarcoplasm
- Disrupted structure: **Ca^{2+} pumps** in the SR membrane
- Consequence: When stimulation stops, ACh is degraded and no new action potentials are generated — but Ca^{2+} cannot be pumped back into SR → Ca^{2+} remains elevated in sarcoplasm → troponin remains bound to Ca^{2+}→ tropomyosin stays displaced → myosin binding sites remain exposed → cross-bridge cycling continues → muscle cannot relax → sustained, uncontrolled contraction

Diagnostic Case Studies

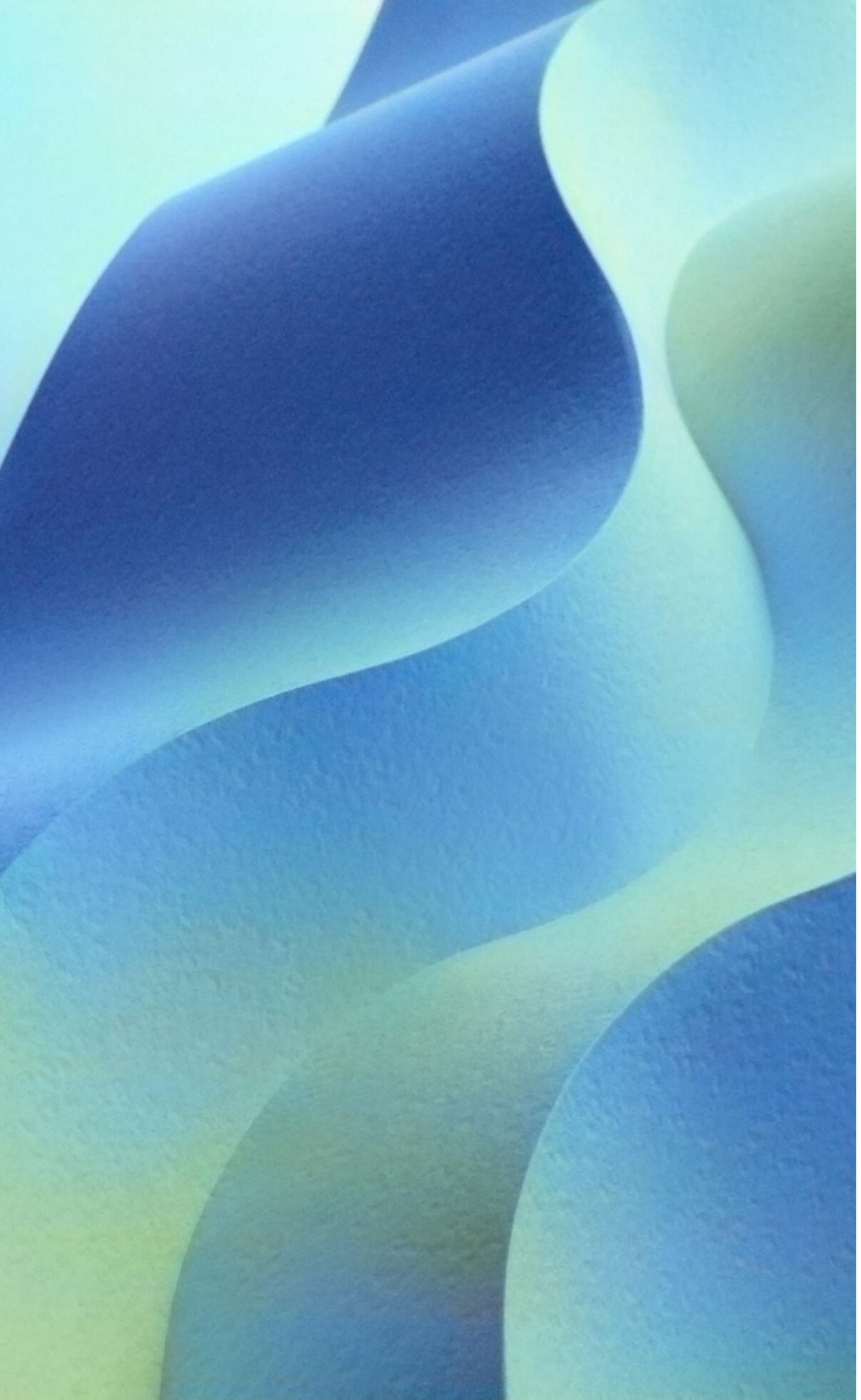
Case 3: A patient has a condition in which their muscles are chronically shortened — sarcomeres are compressed far below optimal resting length. Their physical therapist notes that even with full neural stimulation, the muscles produce surprisingly weak force. Using the length-tension relationship, explain why.

- Failure point: **Contraction phase**
- Disrupted structure: Sarcomere and the **length-tension relationship** — sarcomeres operating far below optimal length
- Explanation: At very short sarcomere lengths, thin filaments from opposite sides of the sarcomere overlap each other and thick filaments begin to compress against Z lines → cross-bridge geometry is disrupted → fewer functional cross-bridges can form simultaneously → tension production is significantly reduced despite maximum neural stimulation

Diagnostic Case Studies

Case 4: A toxin prevents acetylcholinesterase from functioning in the synaptic cleft. ACh is released normally. Predict the consequence for muscle function and explain your reasoning.

- Failure point: **Relaxation phase**
- Disrupted structure: **Acetylcholinesterase** (normally degrades ACh in the synaptic cleft)
- Consequence: ACh accumulates in the synaptic cleft and continues to stimulate ACh receptors repeatedly → motor end plate is continuously depolarized → unrelenting action potentials generated in sarcolemma → continuous Ca^{2+} release → sustained, uncontrolled muscle contraction → eventually the motor end plate becomes desensitized or fatigued and contraction may paradoxically fail



Post-Lecture Quiz

Test what you *learned*!

10 minutes for second attempt

Password = **titin**

No notes – just what is stored in your brain